

Project GEWN — Product Requirements Document

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— Core Team

Table of Contents

1. [Executive Summary](#)
 2. [Problem Statement](#)
 3. [Product Vision](#)
 4. [Goals and Objectives](#)
 5. [Target Audience](#)
 6. [User Pain Points](#)
 7. [Feature Specifications](#)
 8. [Functional Requirements](#)
 9. [Non-Functional Requirements](#)
 10. [User Flows and Scenarios](#)
 11. [AI Behavior and Safety Model](#)
 12. [Security Architecture](#)
 13. [MVP Scope](#)
 14. [Development Roadmap](#)
 15. [Technical Architecture](#)
 16. [Risk Analysis](#)
 17. [Success Metrics](#)
 18. [Future Scalability](#)
 19. [Constraints](#)
 20. [Glossary](#)
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1. Executive Summary

Project GEWN is an AI-native desktop copilot platform that operates as an intelligent layer directly above the user's operating system. It combines conversational AI, real-time screen understanding, desktop and browser automation, persistent semantic memory, and a multi-agent orchestration framework into a single unified companion.

Unlike existing AI assistants that operate in isolated browser tabs or standalone chat applications, GEWN is embedded in the desktop environment itself — always available, always contextually aware, and capable of taking meaningful action across the entire computing surface.

GEWN is designed for developers, knowledge workers, students, and power users who spend the majority of their day inside complex workflows and need an intelligent collaborator that understands what they're working on and can act on their behalf.

The platform is engineered for phased delivery across seven discrete, independently shippable milestones — from a functional MVP (floating overlay + conversational AI + basic automation) through to a full autonomous multi-agent ecosystem with a plugin marketplace and cross-device synchronization.

2. Problem Statement

2.1 The Core Gap

Modern personal computing remains fundamentally manual. Despite decades of productivity tooling and a recent surge in AI chatbot adoption, the relationship between user and operating system has not meaningfully changed. The computer waits. The user operates.

Users today:

- Manually switch between five or more applications to complete a single workflow
- Repeat the same setup routines and file organization tasks daily
- Retrieve information by remembering where they stored it rather than what it contains
- Copy and paste content across tools that have zero awareness of each other
- Lose cognitive bandwidth navigating the operating system instead of focusing on actual work

2.2 Why Existing AI Tools Fall Short

Current AI assistants — including frontier models accessed via web chat — fail to address this gap because they are architecturally disconnected from the computing environment they're supposed to assist with:

- They operate inside isolated browser windows with no awareness of what else is on screen
- They cannot read files, launch applications, execute commands, or interact with the OS in any meaningful way
- They have no persistent memory of the user across sessions; every conversation starts cold
- They respond only when explicitly prompted — they cannot observe, detect, or proactively surface relevant assistance
- They offer no workflow automation; the user must translate AI output into manual action themselves

2.3 The Opportunity

There is no production-ready, AI-native operating layer that:

- Understands what is on the user's screen in real time
- Maintains persistent, evolving knowledge of the user's preferences, workflows, and context
- Can take action across the file system, applications, browser, and system settings
- Coordinates multiple specialized AI agents to handle complex, multi-step goals autonomously
- Does all of this within a cohesive, safe, and transparent user experience

GEWN is built to fill this gap.

3. Product Vision

3.1 Vision Statement

To build the world's first truly ambient AI operating companion — a system that understands what you're working on, remembers how you work, and acts intelligently on your behalf, so that the computer adapts to the human rather than the other way around.

3.2 Design Philosophy

GEWN is not a chatbot with desktop access. It is designed from first principles as an operating companion — a system whose primary context is the user's entire computing environment, not a single conversation thread.

The system is built around four core beliefs:

AI should be ambient. It should exist at the edges of awareness — available instantly, but not demanding attention. GEWN floats above the desktop, always reachable, never in the way.

AI should be contextual. Effective assistance requires understanding what the user is doing *right now*, not just what they've typed in a text box. GEWN reads the screen, understands applications, and infers context continuously.

AI should be proactive. The best assistant doesn't wait to be asked for things it already knows you need. GEWN surfaces relevant help, detects errors, and flags opportunities without requiring explicit prompts.

AI should act, not just advise. The last mile of AI usefulness is execution. GEWN closes the loop between recommendation and action — it doesn't just tell you how to organize your downloads folder; it organizes it.

3.3 Long-Term North Star

The long-term vision for GEWN is a fully AI-native computing ecosystem in which the operating system itself becomes intelligent and adaptive. Future versions envision:

- Autonomous background agents that manage research, scheduling, and file organization without interrupting the user
 - Multimodal reasoning across screen content, voice, documents, and real-time data
 - A personalized AI environment that evolves continuously with the user's habits and preferences
 - An open plugin ecosystem through which developers extend GEWN into specialized domains
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4. Goals and Objectives

Goal 1 — Ambient Desktop Presence

Deliver a persistent, always-available AI assistant that integrates directly into the desktop environment via a floating overlay, accessible in under two seconds via keyboard shortcut or wake word from any application.

Goal 2 — Real-Time Context Awareness

Enable GEWN to understand the user's current application state, screen content, open files, and workflow context — providing assistance that is situationally relevant rather than generically prompted.

Goal 3 — Actionable Automation

Allow users to automate repetitive file management, application control, and multi-step workflows using natural language — with safe, tiered execution and full transparency into what actions are being taken.

Goal 4 — Intelligent Assistance Across Domains

Provide proactive, context-injected AI assistance across coding, research, studying, writing, and general productivity workflows — with specialized behavior per domain.

Goal 5 — Persistent Semantic Memory

Build a three-layer memory architecture (session, long-term semantic, workflow pattern) that evolves with the user and allows GEWN to provide increasingly personalized assistance over time.

Goal 6 — Autonomous Multi-Agent Orchestration

Develop a multi-agent framework of specialized AI agents (Coding, Research, Automation, Memory, Scheduling) capable of decomposing complex goals into executable plans and running extended workflows with user oversight.

Goal 7 — Trust Through Transparency

Ensure every action GEWN takes — file operations, terminal commands, API calls — is visible, explainable, and reversible where possible, with a four-tier permission system that gives users complete control without friction.

5. Target Audience

5.1 Primary Audiences

Developers and AI Engineers Software engineers, AI/ML practitioners, and open-source contributors who spend most of their working day in code editors, terminals, and documentation. They have complex, multi-application workflows, benefit significantly from context-aware coding assistance and automation, and are tolerant of advanced configuration in exchange for power.

Primary use cases: Inline error detection and fix suggestion, terminal command automation, codebase navigation, documentation Q&A via RAG, workspace setup automation.

Students and Researchers Computer science students, academic researchers, and learning-focused users who regularly consume, synthesize, and produce large volumes of written content. They benefit from document intelligence, note generation, research assistance, and semantic memory that tracks ongoing projects.

Primary use cases: PDF summarization and Q&A, research synthesis, study workflow management, semantic search across notes.

Knowledge Workers Analysts, writers, strategists, and consultants who operate across many tools simultaneously. They need to move information across platforms efficiently, reduce context-switching overhead, and maintain organized knowledge bases.

Primary use cases: Cross-application workflow automation, intelligent file organization, meeting prep automation, information retrieval.

Creative Professionals Designers, video editors, and digital creators whose workflows span complex, multi-application creative suites. They benefit from workspace automation, reference

material retrieval, and AI assistance integrated directly into their environment.

Primary use cases: Asset organization, reference retrieval via RAG, workspace launch automation.

5.2 Secondary Audiences

Power Users and Automation Enthusiasts Users who heavily customize their computing environments and are comfortable building personal automation systems. They represent early adopters and are likely to engage with the plugin SDK and workflow scripting.

AI-Forward Early Adopters Users motivated by cutting-edge AI systems who want to experience the frontier of autonomous AI interaction. They generate valuable feedback during early phases and drive word-of-mouth growth.

6. User Pain Points

Pain Point	Current Reality	GEWN's Resolution
Workflow Fragmentation	Users manually switch between 5–10 applications per complex task	GEWN understands cross-application context and automates transitions and handoffs
Repetitive Task Overhead	Daily setup routines, file organization, and copy-paste workflows consume hours per week	Natural language workflow automation executes these at instruction
Context-Blind AI	Existing AI assistants have no awareness of screen content, open files, or application state	Real-time screen capture, OCR, and multimodal analysis give GEWN full situational awareness
Zero Memory Continuity	Every AI conversation starts cold; users re-explain context repeatedly	Persistent three-layer memory system maintains continuity across sessions and evolves over time
Advice Without Action	AI outputs require the user to manually implement every recommendation	GEWN closes the execution loop — it acts, with tiered permission safety
Cognitive Load of Navigation	Mental energy spent managing applications rather than doing actual work	GEWN absorbs operational overhead, letting the user focus on the work itself
File Discovery Friction	Files are found by remembering their location rather than their	Semantic tagging and vector-indexed file retrieval find files by

Pain Point	Current Reality	GEWN's Resolution
	content	meaning
Fragmented Integrations	Productivity tools (Notion, GitHub, Gmail, Slack) don't talk to each other	GEWN's plugin system bridges these as first-class integrations with a unified AI interface

7. Feature Specifications

7.1 Floating Desktop Overlay

The primary user interface is a transparent glassmorphism overlay rendered via Electron's always-on-top transparent window, built in React with Framer Motion animations.

Interface Properties

- Always-on-top rendering above all desktop applications
- Draggable to any screen position; position persists across sessions
- Resizable panel with minimize-to-icon mode
- Configurable opacity and size
- Instant focus via configurable global keyboard shortcut (default: `Ctrl+Space`)
- Smooth animated transitions between states (expanded, minimized, voice mode)
- Futuristic dark UI theme with glassmorphism aesthetics

Interaction Modes

Mode	Description	Activation
Standard	General conversation and assistance	Default
Focus	Minimized footprint; GEWN suppresses unsolicited suggestions	Manual toggle or smart detection
Research	Document analysis and structured output optimized	Contextual or manual
Automation	Step-by-step workflow execution with inline confirmation	Triggered by automation requests
Voice	Full-screen voice visualization; real-time conversation	Wake word or hotkey hold

7.2 Conversational AI Engine

Core Capabilities

- Multi-provider LLM integration (OpenAI, Google Gemini) with hot-swappable model selection
- Streaming response rendering for perceived low latency
- Full conversation history maintained per session and optionally persisted across sessions
- Context injection from: current screen state, active application, long-term memory, RAG retrieval, and workflow state
- Markdown and syntax-highlighted code rendering in the chat panel
- Follow-up understanding and multi-turn reasoning within session context
- Domain-aware prompt orchestration (coding mode, research mode, productivity mode)

AI Behavioral Standards *(see Section 11 for full specification)*

- Leads with direct, capable responses — avoids excessive hedging or filler
 - Transparently surfaces uncertainty; never fabricates file system or screen state
 - Respects scope boundaries; does not expand to unasked actions without surfacing the opportunity first
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7.3 Desktop Automation

All automation actions are classified by risk tier and handled accordingly (see Section 11.1).

File System Operations

- Create, rename, move, copy, and delete files and folders
- Batch operations across directory trees
- Semantic file tagging for content-aware organization
- Intelligent categorization and organization of directories by file type, date, or semantic content
- Natural language file search (e.g., "find the Q3 report I was editing last week")

Application Control

- Launch, focus, and close applications by name or category
- Window arrangement and multi-monitor workspace configuration

- Application-specific workflow presets (e.g., "prepare my coding workspace")

System Utilities

- Clipboard monitoring and intelligent clipboard history
- Screenshot capture (on-demand and periodic for context)
- System notifications and smart reminders
- Global shortcut registration
- Terminal command execution with safety validation and argument sandboxing

Browser Automation (*Playwright-powered*)

- Web navigation, content extraction, and structured summarization
 - Form interaction and submission (with T3 approval)
 - Multi-tab research workflows
 - File download management
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7.4 Screen Awareness and Vision System

Capture and Analysis Pipeline

- On-demand screenshot capture via keyboard shortcut or voice command
- Passive periodic capture for proactive context awareness (configurable, opt-in)
- PaddleOCR primary / Tesseract fallback for text extraction from screenshots
- Multimodal LLM analysis for UI understanding, layout comprehension, and semantic content interpretation

Contextual Assistance

- Real-time coding error detection from screen content
- UI/UX analysis and improvement suggestion from visual inspection
- Document and form understanding from screen capture
- "What is this?" quick-explain overlay for unfamiliar UI elements

Visual Overlay System

- Transparent annotation layer rendered above all content using Electron's overlay window

- Step-by-step interactive guidance rendered directly on screen
- Highlight and annotation of referenced screen elements during AI explanation

7.5 Voice Interaction System

Pipeline Architecture

Wake Word (Porcupine) → STT (Whisper local / Deepgram cloud) → LLM → TTS (ElevenLabs / Coqui)
↑ WebSocket audio stream ↑

Capabilities

- Wake word activation ("Hey GEWN") with configurable sensitivity
- Hold-to-talk alternative for privacy-preferred environments
- High-accuracy continuous speech-to-text with end-of-speech detection
- Interruption handling during TTS playback
- Emotion-aware text-to-speech with natural cadence
- Sub-1.5-second perceived response latency target
- Local STT fallback (Whisper) for Privacy Mode operation

7.6 AI Memory System

Architecture: Three-Layer Memory

Layer	Technology	Scope	Description
Short-Term	Redis	Session	Current conversation state, recent screen context, active workflow state
Long-Term Semantic	PostgreSQL + ChromaDB	Persistent	User preferences, interaction history, learned behavioral patterns, stored knowledge
Workflow Memory	ChromaDB	Task-Specific	Recurring task patterns, workflow templates derived from observed user behavior

Memory Transparency and Control

- Users can query what GEWN remembers at any time ("What do you know about me?")

- All stored memories are viewable, editable, and individually deletable via the Memory Panel
- GEWN never stores credentials, passwords, or private keys in the memory system
- Memory contents are encrypted at rest (AES-256-GCM)

Second Brain Mode

- Explicit knowledge storage: users can instruct GEWN to remember specific facts, preferences, or project context
 - Semantic retrieval: stored knowledge is surfaced contextually during relevant conversations
 - Manual knowledge management interface for reviewing and curating stored information
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7.7 RAG — Document Intelligence

Ingestion Pipeline

- PDF, Markdown, plain text, and rich text document upload
- Chunking, embedding generation, and vector indexing via ChromaDB
- Automatic document summarization on ingestion

Retrieval and Interaction

- Natural language Q&A against uploaded document corpus
 - Semantic search returning ranked, source-attributed excerpts
 - Cross-document synthesis for research workflows
 - Context-aware injection: retrieved content automatically enriches AI reasoning context for related queries
 - Note and insight generation from document analysis
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7.8 Autonomous AI Agent Framework

Agent Roster

Agent	Responsibility
Coding Agent	Code review, error explanation, fix generation, terminal interaction
Research Agent	Web browsing, document synthesis, structured output generation

Agent	Responsibility
Automation Agent	File system operations, application control, workflow execution
Memory Agent	Memory retrieval, storage, pattern extraction, knowledge management
Scheduling Agent	Calendar integration, reminder creation, time-based workflow triggers

Orchestration Model

- LangGraph-powered multi-agent coordination with shared context store
- Planning engine decomposes complex goals into ordered, dependency-aware task graphs
- Agents run sequentially or in parallel based on dependency structure
- Background execution with live progress visibility in the overlay
- User cancellation available at every checkpoint
- Full execution log persisted per workflow run

Safety Constraints on Agents

- Agents cannot initiate T3/T4 actions without explicit in-session user approval
- Scope enforcement: agents do not expand beyond the boundaries of the user's stated goal
- Failed steps surface clearly with options to retry, skip, or abort — no silent continuation

7.9 Plugin System (*Phase 6*)

- Python SDK for third-party developers to build custom commands, agents, workflows, and API integrations
- Plugins run in isolated subprocesses with declared permission scopes and resource caps (25% CPU, 512MB RAM)
- No direct access to GEWN's internal memory or the broader file system outside a plugin-specific sandbox directory
- Plugins are code-signed against the GEWN plugin registry

Core Built-in Integrations

GitHub · Notion · Gmail · Google Calendar · Slack · Discord · Linear

8. Functional Requirements

8.1 User Interface

- FR-UI-01: GEWN overlay must render above all applications on all active displays at all times
- FR-UI-02: Overlay must be draggable, resizable, and support minimize-to-icon state
- FR-UI-03: A configurable global keyboard shortcut must focus the overlay from any application within 200ms
- FR-UI-04: Chat panel must support streaming response rendering, Markdown, and syntax-highlighted code blocks
- FR-UI-05: Dark futuristic UI theme must be applied consistently; a high-contrast accessibility mode must be available
- FR-UI-06: Interaction mode switching (Standard / Focus / Research / Automation / Voice) must be accessible from the overlay header

8.2 Authentication and Session

- FR-AUTH-01: Secure user login and registration with session token management
- FR-AUTH-02: Session persistence across application restarts
- FR-AUTH-03: All credentials and API keys stored in OS-native secure storage (Keychain / Credential Manager / Secret Service)
- FR-AUTH-04: Local encrypted storage for conversation history and user data using AES-256-GCM

8.3 AI Interaction

- FR-AI-01: LLM integration must support streaming token delivery to the frontend via WebSocket
- FR-AI-02: Context injection pipeline must combine screen state, memory retrieval, and RAG content before each inference call
- FR-AI-03: Model selection must be configurable; the system must support at minimum OpenAI and Google Gemini providers
- FR-AI-04: Conversation history must be maintained per session and optionally persisted across sessions

8.4 Automation

- FR-AUTO-01: All automation actions must be classified into one of four risk tiers before execution
- FR-AUTO-02: T1 actions execute silently; T2 actions display a dismissable toast; T3 actions require explicit approval; T4 actions require typed confirmation with full action preview
- FR-AUTO-03: File path operations must validate against an authorized scope allowlist before execution
- FR-AUTO-04: Terminal commands must be constructed as discrete subprocess argument arrays — never via string interpolation of user input
- FR-AUTO-05: All executed automation actions must be logged with timestamps and outcomes in the tamper-evident audit log

8.5 Vision and Screen

- FR-VIS-01: Screenshot pipeline must support on-demand and configurable periodic capture
- FR-VIS-02: OCR extraction must produce text output within 2 seconds for standard screen content
- FR-VIS-03: Multimodal LLM analysis must identify UI elements, code, and document content from screen captures
- FR-VIS-04: Visual overlay system must render transparent annotations above all applications without interfering with user interaction

8.6 Voice

- FR-VOICE-01: Wake word detection must operate with configurable sensitivity and less than 300ms activation latency
- FR-VOICE-02: STT pipeline must support both local (Whisper) and cloud (Deepgram) processing, switchable at runtime
- FR-VOICE-03: End-to-end voice response latency (wake word → TTS playback start) must not exceed 1.5 seconds under normal conditions
- FR-VOICE-04: Voice pipeline must handle speaker interruption gracefully

8.7 Memory

- FR-MEM-01: Short-term memory must persist for the duration of each session and expire on close unless explicitly promoted
- FR-MEM-02: Long-term memory must persist across sessions and be semantically indexed for retrieval
- FR-MEM-03: Users must be able to view, edit, and delete any stored memory via the Memory Panel
- FR-MEM-04: GEWN must never store credentials, passwords, or private keys in any memory layer

8.8 RAG

- FR-RAG-01: Document ingestion must support PDF, Markdown, and plain text formats
- FR-RAG-02: Embedding generation and vector indexing must complete within 30 seconds for documents up to 100 pages
- FR-RAG-03: Semantic search must return source-attributed results with relevance scores
- FR-RAG-04: Retrieved content must be injected into the AI context window automatically for related queries

8.9 Agent System

- FR-AGENT-01: Planning engine must decompose multi-step goals into ordered task graphs before execution begins
- FR-AGENT-02: Agents must share state via a common context store accessible to all active agents in a workflow
- FR-AGENT-03: Background workflow execution must display live progress in the overlay with a cancel control at every checkpoint
- FR-AGENT-04: Failed agent steps must surface actionable options (retry / skip / abort) — silent continuation is not permitted

9. Non-Functional Requirements

9.1 Performance

Metric	Target
Overlay focus latency (keyboard shortcut)	< 200ms
First token rendered (LLM streaming)	< 800ms
Wake word → TTS start	< 1.5 seconds
OCR extraction (standard screen)	< 2 seconds
Vector retrieval (semantic search)	< 500ms
Document ingestion (100 pages)	< 30 seconds

9.2 Reliability

- Graceful degradation when AI providers are unreachable (cached responses, local model fallback)
- Automation safety checks prevent partial execution from leaving the file system in an undefined state
- Agent workflows support checkpoint-based rollback for reversible action sequences
- All T3/T4 actions require explicit approval before execution — no automatic retry on failure

9.3 Security

- All external API communication encrypted via TLS 1.3
- Credentials and API keys stored exclusively in OS-native secure enclaves
- All secrets absent from application files, logs, and memory system
- File system operations validated against an authorized scope allowlist; path traversal blocked by design
- Sensitive file patterns (.env, private keys, SSH credentials, browser credential stores) blocked from all operations
- Plugin execution sandboxed with resource limits and no internal system access

9.4 Privacy

- Privacy Mode routes all processing to local models (Ollama) and local storage; zero external data transmission
- Persistent visual indicator confirms Privacy Mode active state

- Screen capture and OCR data never persisted beyond the immediate inference call unless explicitly instructed by the user
- Prompt injection defense: external content always marked as [EXTERNAL_CONTENT] and architecturally separated from system instructions

9.5 Scalability

- Modular service architecture with clean boundaries between frontend, API gateway, AI engine, automation layer, vision module, voice pipeline, and memory system
- Each module independently deployable and replaceable
- Agent framework extensible via new agent definitions without core system modification
- Plugin system allows capability extension without modifying the core codebase

9.6 Usability

- All critical functions reachable within two interactions from any state
- Permission approval UX designed to be safe without being obstructive
- Reversible actions offer an undo window within the active session
- All AI reasoning and action decisions explainable on request

9.7 Maintainability

- Clear module boundaries documented in ARCHITECTURE.md
- Backend services containerized via Docker for consistent local and production environments
- CI/CD via GitHub Actions for automated testing and build validation
- Audit log is tamper-evident, append-only, user-owned, and exportable

10. User Flows and Scenarios

Scenario 1 — Real-Time Coding Assistance

User: A software engineer encounters a runtime error in VS Code.

Flow:

1. GEWN's passive screen capture detects an exception stack trace in the active window
 2. GEWN proactively surfaces a toast: *"Looks like a TypeError on line 47 — want me to explain it?"*
 3. User clicks the toast or says "Yes"
 4. GEWN analyzes the stack trace via multimodal LLM, explains the root cause in plain language
 5. GEWN offers a corrected code snippet with diff highlighting
 6. User clicks "Apply Fix" → GEWN copies the corrected code to clipboard with a confirmation toast
 7. Full interaction logged; memory system stores that the user is working on this codebase
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Scenario 2 — Natural Language Workflow Automation

User: "Organize my downloads folder."

Flow:

1. GEWN enters Automation Mode, displays a planning step: *"I'll scan Downloads, categorize by type, and create subfolders. Want me to show you the plan first?"*
 2. User confirms
 3. GEWN displays a structured plan: folders to be created, categorization logic, estimated file count per category
 4. User approves (T2 action — create folders) via single click
 5. GEWN executes: creates subdirectories, moves files, reports summary ("Moved 47 files across 6 categories")
 6. Reversible undo option displayed for 30 seconds post-completion
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Scenario 3 — Research and Document Intelligence

User: Uploads three research PDFs, asks: "What are the main arguments across these papers?"

Flow:

1. GEWN ingests all three PDFs: chunks, embeds, indexes in ChromaDB
2. Research Agent performs cross-document synthesis via multi-source RAG retrieval

3. GEWN returns a structured comparative summary with source attribution per claim
 4. User asks follow-up: "Which author disagrees with the second paper's methodology?"
 5. GEWN retrieves relevant passages, identifies the disagreement, surfaces attributed quotes
 6. User requests: "Generate study notes from all three" → GEWN produces a structured Markdown note file
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Scenario 4 — Workspace Automation

User: "Set up my morning coding workspace."

Flow:

1. GEWN references workflow memory: user's coding workspace preset (VS Code + Chrome tabs + Notion)
 2. If no preset exists, GEWN asks: *"What applications and tabs should I open?"* and stores the answer
 3. T1 actions (open apps) execute silently
 4. GEWN reports completion: *"Your workspace is ready — VS Code, Chrome with your GitHub dashboard, and Notion are open."*
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Scenario 5 — Autonomous Research Agent Workflow

User: "Research the top 5 AI agent frameworks and summarize their trade-offs."

Flow:

1. Planning engine decomposes the goal into a task graph: [search for frameworks] → [fetch each documentation page] → [compare features] → [synthesize report]
 2. Research Agent executes each step via browser automation with live progress displayed
 3. At each consequential step (form interaction, downloads), user approval is requested
 4. GEWN compiles a structured comparison report and offers to save it as a Markdown file
 5. Full workflow run persisted in audit log
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Scenario 6 — Proactive Memory Assistance

User: Returns to work after a week away and opens their project folder.

Flow:

- 1. GEWN detects the active application context (VS Code, known project path)
- 2. GEWN cross-references long-term memory: last session context, open issues, incomplete tasks
- 3. GEWN surfaces a non-intrusive context card: *"Welcome back — last week you were working on the auth module. You left a TODO on line 142. Want to pick up there?"*
- 4. User clicks "Yes" → GEWN navigates to the relevant line in VS Code

11. AI Behavior and Safety Model

11.1 Risk Tier Classification

Every action GEWN takes is classified into one of four tiers before execution. The tier governs how approval is handled:

Tier	Examples	User Experience
T1 — Low Risk	Read files, open apps, summarize screen content, web browsing	Execute silently; brief status in overlay
T2 — Medium Risk	Create files/folders, rename, move items	Dismissable toast with undo option
T3 — High Risk	Delete files, run terminal commands, send web form data	Blocking confirmation dialog with full action preview; required every time
T4 — Critical Risk	Modify system settings, send external communications, access sensitive paths	Typed confirmation required + complete action preview

11.2 Core Behavioral Principles

Capable and Direct — GEWN leads with answers and actions. It does not over-explain, hedge unnecessarily, or pad responses with disclaimers. It is explicitly honest about the limits of its knowledge.

Scope Enforcement — GEWN does not expand beyond what the user requested. If it identifies an opportunity outside the task scope, it surfaces the observation as a suggestion and waits for explicit authorization before acting.

Hallucination Prevention — GEWN never generates factual claims about the user's files, system state, or external services without first reading them directly. Uncertainty is surfaced as uncertainty — not filled with plausible-sounding inference.

Honest Failure Handling — When a tool fails or a workflow partially completes, GEWN reports clearly: what succeeded, what failed, and what the available options are. It does not automatically retry T3/T4 actions.

Memory Transparency — GEWN answers accurately and completely when asked what it remembers. It never stores sensitive personal or credential data in the memory system.

Prompt Injection Defense — All external content (documents, web pages, OCR output, clipboard) is injected into the AI context as clearly marked [EXTERNAL_CONTENT] data, architecturally isolated from system instructions. Output validation confirms that inferred actions align with the original user request before execution proceeds.

12. Security Architecture

12.1 Threat Model

GEWN operates with screen access, file system control, terminal execution capability, browser automation, and AI reasoning — a high-privilege attack surface that requires defense-in-depth. The security model is designed to contain compromise at every layer.

12.2 Key Security Controls

Principle of Least Privilege — Every module starts with zero permissions. Access is granted only for the specific resource and minimum duration required.

File System Safety — File paths are canonicalized and validated against an authorized scope allowlist before any operation. Path traversal via `../`, symlinks, and encoded separators is blocked

by design. Sensitive patterns (private keys, .env, SSH credentials, browser credential stores) are blocked from all GEWN operations.

Command Execution Safety — Terminal commands are never constructed via string interpolation of user input. Arguments are always passed as discrete subprocess parameters. An allowlist governs which command categories GEWN can invoke without elevated confirmation.

API Key Security — All secrets are stored in OS-native secure storage (macOS Keychain, Windows Credential Manager, Linux Secret Service). Keys are never written to application files, logs, or the memory system. All API communication uses TLS 1.3.

Local Encryption — Long-term memory, conversation history, and user data are encrypted at rest using AES-256-GCM. The encryption master key is stored in the OS secure enclave and never written to disk in plaintext.

Plugin Sandboxing — Plugins execute in isolated subprocesses with declared permission scopes, resource limits (25% CPU, 512MB RAM), and no direct access to GEWN's internal memory or the broader file system outside a plugin-specific directory. Plugins are code-signed against the GEWN plugin registry.

Privacy Mode — When activated, all processing routes exclusively to local models and local storage. Zero data is transmitted to external services. A persistent visual indicator confirms Privacy Mode active status.

Audit Log — A tamper-evident, append-only log records all consequential actions (file operations, terminal commands, API calls, permission grants) with timestamps and outcomes. The log is user-owned and exportable at any time.

13. MVP Scope

13.1 Included in MVP (Phase 1)

The MVP delivers a functional, useful product that establishes the core value proposition: a floating AI assistant that understands the desktop, can act on it safely, and remembers the user across sessions.

Floating Overlay UI

- Draggable, resizable transparent overlay
- Dark futuristic glassmorphism UI theme
- Chat interface with streaming, Markdown, and code rendering
- Global keyboard shortcut activation
- Minimize-to-icon mode

Conversational AI

- Multi-provider LLM integration (OpenAI, Gemini)
- Streaming responses via WebSocket
- Session conversation history
- Long-term memory with basic semantic retrieval
- Context injection from screen state and memory

Desktop Automation

- Open, close, and switch applications
- Create, rename, move, and organize files and folders
- Four-tier permission system with tiered approval UX
- Basic terminal command execution (T3-gated)

Screen Awareness

- On-demand screenshot capture
- OCR text extraction from screen
- Multimodal LLM analysis of screen content
- Contextual assistance based on detected application state

Voice Interaction

- Wake word detection
- Speech-to-text (Whisper local + Deepgram cloud)
- Text-to-speech (ElevenLabs cloud + Coqui local)
- Basic real-time voice conversation mode

Document Intelligence (Basic RAG)

- PDF and text document upload
- Semantic search and contextual Q&A
- Source-attributed retrieval

Security Baseline

- OS-native credential storage
- AES-256-GCM local encryption
- Audit log (append-only)
- Prompt injection defense
- Authorized file scope enforcement

13.2 Explicitly Excluded from MVP

The following capabilities are architecturally planned but deferred to later phases to maintain MVP focus:

- Advanced multi-agent orchestration and planning engine (Phase 5)
- Browser automation via Playwright (Phase 3)
- Visual overlay annotation system (Phase 4)
- Plugin SDK and marketplace (Phase 6)
- Cross-device sync and mobile companion (Phase 7)
- Local LLM via Ollama for full Privacy Mode (Phase 7)
- Calendar and scheduling integrations (Phase 6)
- Enterprise collaboration features (Phase 7)
- AR/VR interface layer (post-Phase 7)

14. Development Roadmap

GEWN is delivered in seven discrete phases, each independently shippable and each building on stable foundations from the previous phase. No phase begins until its predecessor reaches production stability.

Phase	Focus	Key Deliverables	Timeline Estimate
Phase 1 — MVP Core	Floating overlay, conversational AI, basic memory, file/app control, basic RAG	Working desktop companion with conversation, memory, and safe automation	10–16 weeks
Phase 2 — Voice AI	Full voice pipeline, wake word, STT/TTS, real-time voice mode	Hands-free operation; sub-1.5s end-to-end latency	8–12 weeks
Phase 3 — Desktop Automation	Advanced file management, browser automation (Playwright), workflow presets, task scheduler	Complex cross-application automation; web research workflows	10–14 weeks
Phase 4 — Vision and Context	Full OCR pipeline, screen understanding, coding error detection, visual overlay annotations, proactive assistance	GEWN detects and responds to screen events without prompting	10–14 weeks
Phase 5 — Autonomous Agents	Multi-agent orchestration (LangGraph), planning engine, background task execution, agent coordination	Complex multi-step autonomous goals; background agent workflows	12–16 weeks
Phase 6 — Plugin Ecosystem	Developer SDK, plugin sandboxing, core integrations (GitHub, Notion, Gmail, Slack, Linear, Calendar), plugin registry	Extensible platform; third-party developer support	12–16 weeks
Phase 7 — Multi-Device and Enterprise	Cross-device sync, mobile companion app, local LLM (Ollama), enterprise workspace features	Full offline capability; mobile extension; team features	14–20 weeks

Guiding Principle: Ship something genuinely useful before adding something ambitious. Each phase must produce a better product than the previous phase — not just more features.

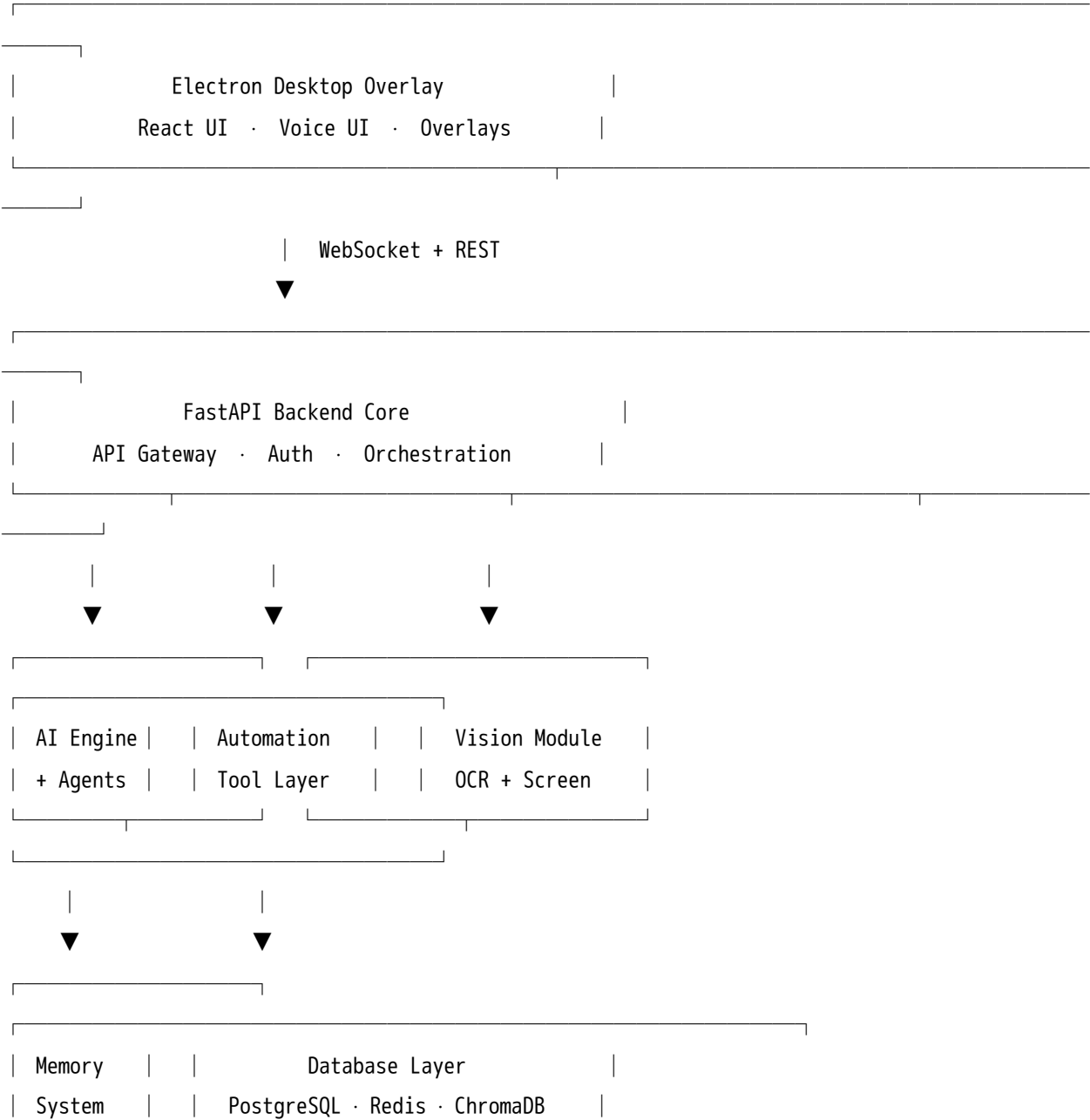
14.1 Phase 1 Milestones

Milestone	Deliverable
M1.1	Electron overlay shell: window launches, floats, drags, minimizes
M1.2	Chat interface with streaming LLM responses via WebSocket
M1.3	Short-term session memory; conversation history persistence
M1.4	Basic file and application automation with T1/T2/T3 permission UX

Milestone	Deliverable
M1.5	On-demand screenshot capture and OCR extraction
M1.6	Long-term semantic memory (ChromaDB) with basic retrieval
M1.7	PDF ingestion and basic RAG Q&A
M1.8	Security baseline: credential storage, encryption, audit log

15. Technical Architecture

15.1 System Overview



15.2 Technology Stack

Layer	Technology	Rationale
Desktop Framework	Electron	Cross-platform overlay with OS-level access and transparent window support
UI Framework	React + Tailwind CSS + Framer Motion	Component-based UI with performant animation and utility-first styling
Backend	Python + FastAPI + WebSockets	High-performance async API with native AI/ML library ecosystem
LLM Providers	OpenAI API, Google Gemini API	Frontier model access; multi-provider for resilience and model selection
Agent Orchestration	LangChain + LangGraph	Tool-calling abstraction and stateful multi-agent coordination
Wake Word	Porcupine	Lightweight on-device wake word detection; low CPU overhead
Speech-to-Text	Whisper (local), Deepgram (cloud)	Local for Privacy Mode; cloud for quality and speed
Text-to-Speech	ElevenLabs (cloud), Coqui TTS (local)	Natural voice quality; local fallback for Privacy Mode
Desktop Automation	PyAutoGUI	Cross-platform OS-level GUI interaction
Browser Automation	Playwright	Modern, reliable browser control with async support
OCR	PaddleOCR (primary), Tesseract (fallback)	High accuracy; open-source; local execution
Vector Database	ChromaDB (local), Pinecone (cloud option)	Semantic embedding storage and retrieval for memory and RAG
Relational Database	PostgreSQL	Structured persistent storage for user data and conversation history
Cache / Session	Redis	Fast in-memory session state and short-term memory
Local LLM	Ollama	Full offline inference for Privacy Mode (Phase 7)
DevOps	Docker + GitHub Actions	Consistent environments; automated CI/CD

15.3 Data Flow — Standard Query

- User Input (text or voice)
- Frontend captures input and current screen state
 - WebSocket sends to FastAPI
 - Memory Agent retrieves relevant long-term context
 - RAG pipeline retrieves relevant document context (if applicable)
 - Context assembled: [system prompt] + [screen state] + [memory] + [RAG] + [user input]
 - LLM inference with streaming
 - Stream tokens back to frontend via WebSocket
 - If action detected: classify tier → request approval → execute → log
-

16. Risk Analysis

16.1 Technical Risks

Risk	Probability	Impact	Mitigation
LLM API latency degrades real-time UX	Medium	High	Streaming rendering; multi-provider fallback; local model option in Privacy Mode
OCR accuracy insufficient for complex UIs	Medium	Medium	Multi-engine fallback (PaddleOCR + Tesseract); multimodal LLM as verification layer
Desktop automation breaks across OS updates	High	Medium	Comprehensive OS compatibility testing; abstraction layer over PyAutoGUI
Memory system context window overflow on long sessions	Medium	Medium	Intelligent memory summarization and compression; tiered eviction policy
Agent orchestration complexity introduces reliability failures	Medium	High	Checkpoint-based execution; explicit failure surfacing; no silent continuation

16.2 Security Risks

Risk	Probability	Impact	Mitigation
Prompt injection via malicious document content	Medium	High	Architectural separation of external content; output validation against original intent
Unauthorized file system access via crafted inputs	Low	Critical	Path canonicalization; scope allowlist enforcement; sensitive pattern blocklist

Risk	Probability	Impact	Mitigation
API key exfiltration	Low	Critical	OS-native secure storage only; keys absent from all logs and memory
Plugin executing malicious code	Low	High	Code signing; sandboxed subprocess execution; resource limits

16.3 Privacy Risks

Risk	Probability	Impact	Mitigation
Screen content containing sensitive data sent to cloud LLM	Medium	High	User-configurable screen capture controls; Privacy Mode for full local processing
Memory system storing inadvertently sensitive information	Medium	Medium	Explicit blocklist for credential patterns; full user memory transparency and deletion
Voice pipeline transmitting audio to cloud STT without consent	Low	High	Clear disclosure of STT provider; local Whisper option; no audio retention

16.4 Product and Business Risks

Risk	Probability	Impact	Mitigation
AI hallucinations cause incorrect automation actions	Medium	High	Require confirmation for all file-modifying actions; no silent automation of T3/T4
System performance impact from passive screen monitoring	Medium	Medium	Configurable capture frequency; smart activation only when GEWN is focused
Cross-platform inconsistency (macOS / Windows / Linux)	High	Medium	Platform abstraction layers; phased OS support (macOS first, then Windows)
User trust loss from unexpected autonomous action	Low	Critical	Strict permission tier enforcement; comprehensive audit log; undo windows

17. Success Metrics

17.1 Product Health Metrics

Metric	Description	Target (90 days post-launch)
Daily Active Users	Unique users opening GEWN on a given day	Growth trend; absolute target TBD by market size
Session Duration	Average time GEWN is actively used per day	> 20 minutes
Day-7 Retention	% of new users returning after 7 days	> 40%
Day-30 Retention	% of new users returning after 30 days	> 25%
Automation Adoption	% of MAU who execute at least one automation per week	> 50%
Memory Utilization	% of MAU whose long-term memory contains > 10 stored facts	> 60%

17.2 AI Quality Metrics

Metric	Description	Target
Task Completion Rate	% of user-initiated tasks that reach the user's intended outcome	> 80%
Automation Success Rate	% of automation workflows that complete without user-initiated abort	> 90%
RAG Retrieval Relevance	User-rated relevance of document Q&A responses (sampled)	> 4.0 / 5.0
Context Accuracy	% of responses that correctly reference current screen or session state	> 85%
Memory Retrieval Quality	% of memory retrievals rated as relevant by users (sampled)	> 4.0 / 5.0

17.3 User Experience Metrics

Metric	Target
Overlay activation latency	< 200ms p95
First token rendered (LLM)	< 800ms p95
Voice end-to-end latency	< 1.5 seconds p95
User satisfaction (post-session survey, sampled)	> 4.2 / 5.0
Permission dialog abandonment rate (T3/T4)	< 10%

17.4 Leading Indicators (Early Validation)

- Number of automation workflows saved as presets (indicates perceived utility)
 - Number of explicit memory store requests per user (indicates trust in memory system)
 - RAG document upload count per user (indicates adoption of document intelligence)
 - Plugin installations per active user (Phase 6 onward)
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18. Future Scalability

18.1 Multi-Agent Ecosystem Expansion

Beyond the five core agents in Phase 5, the agent framework is designed to support an unlimited registry of specialized agents. Planned future agents include a Writing Agent (long-form content generation and editing), a Finance Agent (budget and transaction analysis), and a Health Agent (habit tracking and wellness workflow automation).

18.2 Plugin Marketplace

A developer-facing plugin marketplace enabling third-party builders to extend GEWN with new commands, agents, workflow templates, and API integrations. Revenue model options include revenue sharing on premium plugins and a free tier for open-source contributions.

18.3 Cross-Device Continuity

A mobile companion app (iOS + Android) that extends GEWN's memory and AI context to mobile workflows. Cross-device sync allows memory, conversation history, and workflow state to be available across all user devices. End-to-end encrypted sync via user-controlled cloud storage.

18.4 Local AI Infrastructure

Full offline operation via Ollama-based local LLM inference, eliminating dependency on cloud AI providers for users who require complete data sovereignty or operate in air-gapped environments.

18.5 Enterprise AI Workspace

Team-level GEWN deployment with shared workflow templates, centralized IT policy control, audit log aggregation, and collaborative agent workflows. Enterprise tier supports SSO, custom plugin allowlists, and admin dashboards.

18.6 Adaptive Operating System Layer

Long-term exploration of GEWN as an OS-level capability — integrating with kernel-level APIs, accessibility frameworks, and system event hooks to provide even deeper context awareness and lower-latency automation with reduced screen-scraping dependency.

19. Constraints

19.1 Technical Constraints

- Desktop automation depends on OS-level permission grants that vary by platform and OS version; some capabilities require explicit user permission on macOS (Accessibility, Screen Recording, Full Disk Access)
- LLM inference latency is subject to provider API performance; real-time responsiveness targets cannot be guaranteed under API degradation
- Local resource consumption (CPU, RAM) must be actively managed to avoid impacting the user's primary workflows; passive monitoring must be configurable

19.2 Platform Constraints

- Phase 1 targets macOS as the primary development platform; Windows support follows in Phase 3; Linux community support in Phase 4
- Electron's cross-platform window management has documented inconsistencies between platforms that require platform-specific handling

19.3 Financial Constraints

- LLM API usage costs scale directly with active user count and context window size; cost modeling must be incorporated into pricing design from Phase 1
- Vector database scaling costs (Pinecone cloud option) require monitoring at scale

- Voice pipeline cloud costs (Deepgram STT, ElevenLabs TTS) require usage caps or tiered plans for cost management

19.4 Regulatory Constraints

- Screen content capture and processing may be subject to data protection regulation (GDPR, CCPA) in certain markets; Privacy Mode and local-only processing options are the primary compliance strategy
- Enterprise deployments may require SOC 2 compliance consideration before Phase 7

20. Glossary

Term	Definition
Overlay	The Electron-powered transparent floating window that serves as GEWN's primary interface
Automation Tier	The risk classification (T1–T4) assigned to each action before execution
RAG	Retrieval-Augmented Generation — the pipeline for injecting retrieved document content into LLM reasoning context
Memory Agent	The AI agent responsible for storing, retrieving, and managing all layers of GEWN's memory system
Planning Engine	The LangGraph-powered component that decomposes complex multi-step goals into ordered, executable task graphs
Second Brain Mode	The explicit user-directed knowledge storage mode that allows intentional information storage in long-term memory
Privacy Mode	The operational mode in which all processing routes to local models and local storage; zero external data transmission
Glassmorphism	The frosted-glass UI aesthetic applied to GEWN's overlay: semi-transparent background with blur effect
Context Injection	The process of combining screen state, memory retrieval, and RAG content into the LLM input before each inference call
Scope Enforcement	The behavioral constraint that prevents GEWN from taking actions outside the explicit boundaries of a user request
Audit Log	The tamper-evident, append-only record of all consequential actions taken by GEWN
Plugin Registry	The code-signed distribution system for third-party GEWN plugins

This document is the source of truth for Project GEWN's product requirements. All feature development, architectural decisions, and scope trade-offs should be evaluated against the goals, constraints, and principles defined here. This document is maintained by the GEWN core team and versioned alongside the codebase.

Related documents: ARCHITECTURE.md · FEATURES.md · ROADMAP.md · AI_BEHAVIOR.md · SECURITY_NOTES.md · USER_FLOW.md